

# Responsible AI for Sustainable Mobility: Predicting Bike Sharing Demand

## 1.Introduction



1.Bike-sharing systems promote green transportation, reduce congestion and improve urban sustainability



2.AI can accurately predict bike demand, but models must be transparent, fair and reliable.



3.We develop an Explainable AI framework to predict demand while ensuring sustainability and responsible AI practices.

## 2.Dataset

Seoul Bike Sharing Demand Dataset(Kaggle)

| Name             | Value             |
|------------------|-------------------|
| Records (Hourly) | 8,760             |
| Features         | 13                |
| Target Variable  | Rented Bike Count |

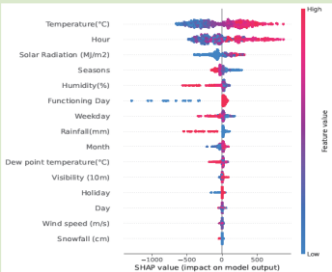
## 3 RESEARCH QUESTIONS

- RQ1**  
Can AI accurately predict bike-sharing demand?
- RQ2**  
Which environmental and temporal factors most influence demand?
- RQ3**  
Can explainable AI improve transparency in predictions?
- RQ4**  
How can AI-based prediction contribute to sustainable urban transportation?
- RQ5**  
What ethical and governance concerns arise from AI-driven mobility prediction systems?

## 4.Evaluation Metrics

| MAE                 | RMSE                   | R <sup>2</sup>  |
|---------------------|------------------------|-----------------|
| Mean Absolute Error | Root Mean Square Error | R-squared Score |

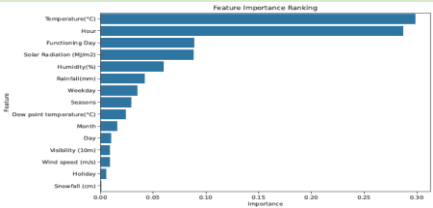
## 8.Explainable AI SHAP Analysis (RQ3)



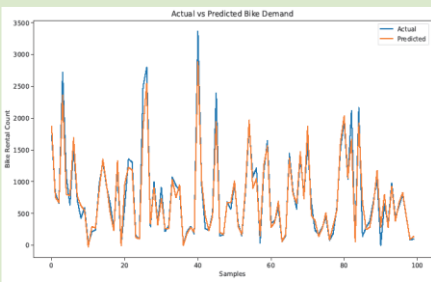
## 5. METHODOLOGY WORKFLOW

- 1. Data Collection**  
Collect Seoul bike-sharing hourly data from Kaggle.
- 2. Data Preprocessing**  
Handle missing values, remove duplicates, encode categorical variables, scale features.
- 3. Exploratory Data Analysis**  
Analyze trends, correlations and distributions to extract insights.
- 4. Feature Engineering**  
Create temporal features (hour, day, month, weekday) and encode variables
- 5. Model Training**  
Train multiple ML models:  
•Linear Regression  
•Decision Tree  
•Random Forest  
•XGBoost
- 6. Explainable AI (SHAP)**  
Use SHAP to explain model predictions and feature contributions.
- 7. Sustainability Analysis**  
Analyze seasonal patterns and estimate environmental benefits.
- 8. Ethical & Governance Analysis**  
Evaluate prediction errors across conditions and identify governance risks.
- 9. Recommendations**  
Provide data-driven and responsible AI recommendations for smart mobility systems.

## 7.Feature Importance(RQ2)



## 6.Model Performance(RQ1)



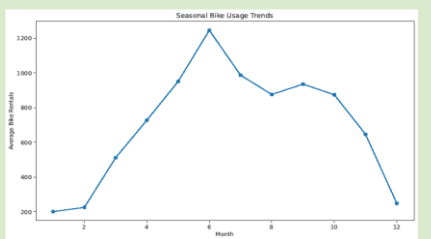
## Model Performance Comparison

| Model   | MAE   | RMSE  | R2 Score |
|---------|-------|-------|----------|
| L.R     | 330.1 | 440.2 | 0.534    |
| D.T     | 131.4 | 254.1 | 0.844    |
| R.F     | 98.5  | 174.3 | 0.927    |
| XGBoost | 98.8  | 168.2 | 0.932    |

## Ethics & Governance (RQ5)

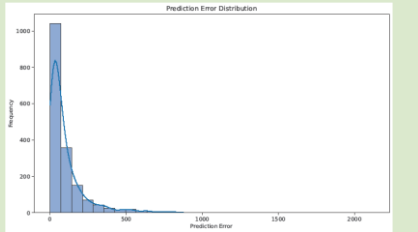
| Risk Area      | Potential Impact                                        | Mitigation Strategy                                |
|----------------|---------------------------------------------------------|----------------------------------------------------|
| Data Bias      | Unequal performance across seasons and holidays         | Regular retraining, bias detection, diverse data   |
| Transparency   | Black-box predictions reduce trust                      | Use XAI (SHAP), provide explanations               |
| Privacy        | User location & usage data may risk privacy             | Data anonymization, secure storage, access control |
| Accountability | Unclear responsibility for wrong predictions            | Human-in-the-loop, clear governance framework      |
| Sustainability | Over-reliance on predictions may lead to resource waste | Sustainable planning, continuous impact assessment |

## 9.Sustainability (RQ4)



## 10.Error Distribution

Higher errors occur during peak hours and holidays. Continuous monitoring and retraining are essential for fair and reliable deployment.



**Conclusion:** XGBoost provided the best performance, with temperature, hour, and humidity having the greatest influence on demand. SHAP improved transparency, demonstrating the value of responsible AI for sustainable mobility.